Exercise sheet: Week 9 - solutions

1. X from $N(\mu, 1)$ and we are testing $H_0 : \mu = -1$ v. $H_1 : \mu = +1$

   a) The Bayesian decision is

$$a_1(\text{do not reject } H_0) \text{ if } \frac{f(x|\mu_0)}{f(x|\mu_1)} > \frac{P(\mu_1)L_{10}}{P(\mu_0)L_{01}} = k_1$$

$$f(x|\mu) \propto \exp\left\{-\frac{1}{2}(x-\mu)^2\right\}$$
$$\therefore \frac{f(x|-1)}{f(x|+1)} = \exp\left\{-\frac{1}{2}[(x+1)^2 - (x-1)^2]\right\}$$
$$= \exp\{-2x\}$$

   $\therefore$ do not reject H_0 if $e^{-2x} > k_1$ i.e. if $-2x > \log k_1$ or $x < -\frac{1}{2}\log k_1$ i.e. $x < k_2$.

   Therefore the Bayesian decision is to not reject H_0 if $x < k$, reject H_0 if $x > k$ and either if $x = k$ (whence $\beta(a_1) = \beta(a_2)$).

   b) Probability of Type I error is $\alpha = \mathbf{P}(a_2 \,|\, \mu = -1)$.

$$\begin{aligned}\therefore \alpha &= \mathbf{P}(X > k \,|\, \mu = -1) \\ &= \mathbf{P}(X + 1 > k + 1) \quad \text{where } Z = X + 1 \sim N(0,1) \\ &= \mathbf{P}(Z > k + 1) \\ &= 1 - \Phi(k+1)\end{aligned}$$

   Probability of Type II error is $\beta = \mathbf{P}(a_1 \,|\, \mu = 1)$.

$$\begin{aligned}\therefore \beta &= \mathbf{P}(X < k \,|\, \mu = +1) \\ &= \mathbf{P}(X - 1 < k - 1) \quad \text{where } Z = X - 1 \sim N(0,1) \\ &= \mathbf{P}(Z < k - 1) \\ &= \Phi(k-1)\end{aligned}$$

   c) $L_{01} = 2, L_{10} = 1$ and $p_0 = \frac{1}{3}, p_1 = \frac{2}{3}$

$$\Longrightarrow k_1 = \frac{1}{2} \times \frac{2/3}{1/3} = 1 \qquad \therefore k_2 = -\frac{1}{2}\log k_1 = 0$$

   So our "k"$= 0$. Therefore

$$\alpha = 1 - \Phi(1) = 1 - 0.8413 = 0.1587$$

and $\beta = \Phi(-1) = 0.1587$.

N-P test with $\alpha = 0.05$:

Also do not reject H_0 if $X < k$, say k^*. k^* is such that $P(z > k^* + 1) = 0.05$. Therefore $k^* + 1 = 1.6449$, i.e. $k^* = 0.6449$, and so

$$\beta = \Phi(k^* - 1) = \Phi(-0.3551) = 1 - 0.639 = \underline{\underline{0.361}}.$$

Note: Bayesian $\alpha >$ N-P α and Bayesian $\beta <$ N-P β.

2. $X \sim$ Bin$(4, p)$ and prior is $p \sim$ Beta$(3, 1)$. We are testing

$$H_0 : p \leq \frac{1}{2} \text{ v. } H_1 : p > \frac{1}{2}$$

a) $p \sim$ Beta$(3, 1)$ so $\pi(p) = 3p^2$, where $0 < p < 1$. It follows that

$$\mathbf{P}(p \leq \frac{1}{2}) = \int_0^{\frac{1}{2}} 3p^2 dp = [p^3]_0^{\frac{1}{2}} = \frac{1}{8}.$$

b) $p|x \sim$ Beta$(x + 3, 4 - x + 1) =$ Beta$(4, 4)$

The posterior density is then $\pi(p|1) = 140p^3(1 - p)^3$. It follows that

$$\mathbf{P}\left(p \leq \frac{1}{2} \,|\, x = 1\right) = \frac{1}{2} \qquad \text{(by symmetry!)}$$

c) The Bayes Principle states that we should select the action a_i in order to minimise the expected posterior loss:

$$B(a_i) = \int \pi(\theta|\underline{x}) L_i(\theta) d\theta.$$

Here we have

$$\pi(p|1) = 140p^3(1 - p)^3,$$

$$L_1(p) = 2\left(p - \frac{1}{2}\right) \quad \text{for} \quad p > \frac{1}{2}$$

and

$$L_2(p) = \left(\frac{1}{2} - p\right) \quad \text{for} \quad p \leq \frac{1}{2}.$$

This gives

$$B(a_1) = \int_{1/2}^{1} 140p^3(1-p)^3 \cdot 2\left(p - \frac{1}{2}\right) dp$$
$$B(a_2) = \int_{0}^{1/2} 140p^3(1-p)^3 \cdot \left(\frac{1}{2} - p\right) dp$$

and we choose a_1 (do not reject H_0) if $B(a_1) - B(a_2) < 0$.

Observe that (taking a factor of -1 outside the second integral):

$$\begin{aligned} B(a_1) - B(a_2) &= \int_{1/2}^{1} 140p^3(1-p)^3 \cdot 2\left(p - \frac{1}{2}\right) dp + \int_{0}^{1/2} 140p^3(1-p)^3 \cdot \left(p - \frac{1}{2}\right) dp \\ &= 2\int_{0}^{1} 140p^3(1-p)^3 \left(p - \frac{1}{2}\right) dp + \int_{0}^{\frac{1}{2}} 140p^3(1-p)^3 \left(\frac{1}{2} - p\right) dp \\ &= 2\left(\mathbf{E}(p \,|\, 1) - \frac{1}{2}\right) + \frac{1}{2}\,\mathbf{P}\left(p \leq \frac{1}{2} \,|\, x = 1\right) - \int_{0}^{\frac{1}{2}} 140p^4(1-p)^3 dp \\ &= 2\left(\frac{1}{2} - \frac{1}{2}\right) + \frac{1}{2} \times \frac{1}{2} - 0.18 = 0.07 \end{aligned}$$

where we had to evaluate

$$\int_{0}^{\frac{1}{2}} 140p^4(1-p)^3 dp = 0.18.$$

Since $B(a_1) - B(a_2) > 0$, the Bayesian decision is to choose $H_1 : p > \frac{1}{2}$.